Egg Segmentation and Tracking

Computer vision internship assignment

Author: Sarang Palsutkar

Code and results: https://github.com/sarangp17/egg-segmentation-tracking

Annotated tracking video (public Google Drive):
https://drive.google.com/file/d/1nzwb-UZDESOID0dqhEN9Ej4_-i6Tr7pG/view?usp=sharing

Summary

I built a pipeline in four stages: (1) pseudo-ground-truth egg masks for all 110 dataset images from the pretrained Segment Anything Model (SAM) plus a selection rule and manual box prompts, (2) a leakage-aware 80:20 split and a classical OpenCV segmentation developed through six logged experiments, (3) an evaluation of the classical masks against the SAM masks on the held-out images, and (4) segmentation and tracking of the eggs in a supplied video.

• Reference masks: 93 of 110 images were produced automatically, 17 of 110 (15.5 percent) needed manual box prompts to SAM.

• Classical method (final, E6): macro IoU 0.496 on the 88 development images and 0.409 on the 22 held-out evaluation images.

• Video: 744 frames (10 s to 41 s) processed. The tracker created 8 IDs for 2 real eggs; IDs are lost when a hand lifts or hides an egg.

Important: all scores measure agreement with SAM-based pseudo-ground truth, not accuracy against human labels. Tables below are generated from the result files saved by the notebook; the narrative numbers are copied from the notebook outputs.

1. Data

• Roboflow Universe dataset fire-bf7qe/egg-vzhtr, version 1, licence CC BY 4.0, 110 JPG images at 640x640 (Roboflow resized them with stretch).

• I exported it in the 'folder' format, which contains images only (no boxes or masks), so SAM masks are the only reference available. Two images sit in an 'Unlabeled' folder; both are boiling eggs in water.

• Roboflow's own train/valid/test folders (77/22/11 images) were ignored. All 110 images were pooled and re-split by me.

• Eggs are white. Backgrounds vary (tray, table cloth, wall, foil, grass, water); a few images have two touching eggs; there are shadows, and the two boiling eggs are translucent.

2. Task 1: pretrained segmentation (SAM)

Model: SAM ViT-B (segment-anything) on a Colab T4 GPU, automatic mask generator (32 points per side, predicted-IoU threshold 0.88, stability threshold 0.92, minimum region 200 pixels).

SAM returns many regions per image (eggs, tray, cloth, background), so I did not treat all proposals as eggs. A region is kept as an egg only if it passes simple checks, then the kept regions are merged and holes are filled:

• size between 2 and 90 percent of the image; solidity (area / convex-hull area) at least 0.88; aspect ratio at most 2.5;

• covers at most 20 percent of the image border (backgrounds wrap around the border, eggs sit inside); mean saturation at most 70; mean brightness at least 170.

• If an image has no passing region, the brightness limit is relaxed to 155 (used for 5 images).

The rule was developed iteratively by looking at overlays. Early versions failed in instructive ways: a rule that dropped regions larger than 45 percent of the image removed the eggs in close-up photos (the egg was region 0); a brightness limit of 155 for every image added false regions (a white panel, a tile, a plastic object); and one-step rules could not separate the translucent boiling eggs from the water.

Manual work (disclosed): 17 of 110 images (15.5 percent) used manual box prompts that I drew after inspecting the images: 12 images use boxes only (boiling eggs, eggs on foil, in grass, held between fingers, an egg on a tray with empty pockets, and others where colour rules gave nothing) and 5 images use the automatic rule plus an extra box for a missed egg. For each box, SAM's mask is cut to the box and only the largest connected piece is kept. Final reference blobs per image: 1 egg in 83 images, 2 in 16, 3 in 11.



Figure 0: eight reference masks (green) chosen with a fixed random seed. The full set of 110 masks and overlays is in the repository results.

3. Task 2: split and classical segmentation

3.1 Leakage-aware split

Each image got a 64-bit difference hash (dHash: shrink to 9x8 grey pixels and compare neighbouring pixels). Images whose hashes differ in at most 10 bits were grouped as near-duplicates (union-find), and whole groups were assigned to the evaluation set with a seeded shuffle (seed 42) until it held about 20 percent of the images.

• Result: 88 development and 22 evaluation images. Shared groups: 0. Shared images: 0.

• The smallest distance between any two images was 12 bits (median 21), so no near-duplicates existed at threshold 10; every image became its own group and the split reduced to a seeded random split. The closest development/evaluation pair differs by 16 bits and the two photos look different.

• Limits of perceptual hashing: it catches copies, resizes and small crops but can miss the same egg photographed from another angle or under other lighting; the threshold is a judgement call.

• Reference-mask methods per split: development 73 automatic, 2 automatic plus box, 11 box only, 2 relaxed-automatic; evaluation 15, 3, 1 and 3 respectively. Boxes were used for 13 of 88 development images and 4 of 22 evaluation images.

• The evaluation images were not used for choosing classical parameters. (The SAM selection rule was inspected on images from the whole dataset before the split, see limitations.)

3.2 Experiment log (development set only)

Each card gives what changed, why I tried it, the parameters, the development-set result, and why I kept or discarded it. E1 to E6 use macro IoU against the SAM masks on the 88 development images. V1 to V3 are the video-only trials (section 5), scored by a proxy IoU on 4 hand-boxed frames. The full log is experiments/experiment_log.csv in the repository.

Trial E1	**Baseline: grayscale + global Otsu threshold, nothing else**
Why tried	Eggs are white, so a bright-pixel threshold on grayscale separates them from the background. Simplest possible baseline to measure everything else against.
Parameters	{'method': 'Otsu', 'blur': 'none'}
Result	macro IoU 0.345, Dice 0.4652, precision 0.3811, recall 0.8706; micro IoU 0.2692 (88 development images)
Decision	keep as baseline, not final
Observations / next	Macro IoU 0.345, precision 0.38, recall 0.87. Finds eggs but also marks bright backgrounds and a finger; images 20 and 44 almost fully missed (egg not bright). Next: E2: HSV threshold (low saturation + high brightness) to drop coloured things like fingers

Trial E2	**Replace grayscale Otsu with HSV rule: saturation <= sat_max AND value >= val_min**
Why tried	Eggs are white: low saturation and high brightness. A colour rule should remove coloured background and fingers that plain brightness keeps. E1 had precision 0.38: it marked fingers and coloured background. Saturation separates white from coloured things.
Parameters	{'sat_max': 40, 'val_min': 170}
Result	macro IoU 0.4241, Dice 0.5465, precision 0.5451, recall 0.6985; micro IoU 0.346 (88 development images)
Decision	keep, current best
Observations / next	Macro IoU 0.424 (up from 0.345), precision 0.55, recall 0.70. Best sat_max=40, val_min=170. Dim or shadowed eggs (images 39, 20, 119, 44) almost fully missed; white panel behind egg (125) and a finger still marked. Next: E3: add morphological opening and closing to remove specks and fill gaps

Trial E3	**Add morphological opening then closing (elliptical kernel) after the E2 colour rule**
Why tried	Opening removes specks and thin bridges (like a finger link); closing fills gaps from dirt and reflections inside eggs. E2 masks are ragged, with holes inside eggs and small false specks in the background.
Parameters	{'sat_max': 40, 'val_min': 170, 'open_k': 9, 'close_k': 0}
Result	macro IoU 0.4468, Dice 0.5709, precision 0.5862, recall 0.6896; micro IoU 0.3679 (88 development images)
Decision	keep, current best
Observations / next	Macro IoU 0.447 (up from 0.424), precision 0.586, recall 0.690. Opening kernel 9 best (largest value tried, so larger not explored); closing hurt in every setting and was dropped. Dim-egg images and white panel image still fail. Next: E4: filter blobs by size, border contact and elongation to drop panels, fingers and specks

Trial E4	**Add connected-component blob filter (min size, border contact, aspect ratio) after E3**
Why tried	Backgrounds and fingers form blobs that are large and wrap the image border, or are long and thin; eggs are compact blobs inside the picture. E3 still marks a white panel (image 125) and a finger; colour alone cannot tell them from eggs, shape and position can.
Parameters	{'min_area_frac': 0.02, 'max_edge': 1.0, 'max_aspect': 2.5}
Result	macro IoU 0.4924, Dice 0.6115, precision 0.6876, recall 0.6603; micro IoU 0.4015 (88 development images)
Decision	keep, current best
Observations / next	Macro IoU 0.492 (up from 0.447), precision 0.688, recall 0.660. Best min_area 2% of image; border-contact filter hurt in every setting (close-up eggs touch the border) so max_edge=1.0 (off). New zero-IoU images 149 and 150 suggest the filter removed their eggs. Next: E5: watershed to split touching eggs before the blob filter

Trial E5	Insert distance-transform watershed (per-blob peaks) between E3 cleaning and the E4 blob filter
Why tried	Touching eggs form one merged blob that fails the shape filter; splitting them with watershed first lets each egg pass. E4 removed some eggs (images 149, 150) likely because merged eggs became one long blob.
Parameters	{'peak_frac': 0.5}
Result	macro IoU 0.442, Dice 0.5593, precision 0.688, recall 0.5433; micro IoU 0.3811 (88 development images)
Decision	discard
Observations / next	Macro IoU 0.442 (worse than E4 0.492), recall dropped to 0.543. Best peak_frac was 0.5, the setting that splits least, suggesting watershed over-splits single eggs and the cut lines remove pixels (not verified). Next: E6: images 149/150 lost eggs because blob aspect ratio 2.7-3.6 failed the 2.5 limit; tune max_aspect

Trial E6	Keep E3 + E4 filter (min area 2%, no border rule); only raise max_aspect
Why tried	Touching eggs form long blobs (aspect 2.7-3.6); the 2.5 limit removed them, so a looser limit recovers them. Diagnosis of images 149 and 150 showed egg blobs removed only by the aspect-ratio check.
Parameters	{'min_area_frac': 0.02, 'max_edge': 1.0, 'max_aspect': 3.0}
Result	macro IoU 0.4963, Dice 0.6181, precision 0.684, recall 0.6772; micro IoU 0.3997 (88 development images)
Decision	keep as FINAL algorithm
Observations / next	Macro IoU 0.496 (E4 was 0.492). max_aspect 3.0 recovered touching-egg blobs (images 149/150). Worst images are dim or shadowed eggs; blue in error maps is mostly egg pixels below the brightness limit. Next: Freeze parameters; run one final evaluation on the held-out evaluation set

How the experiments led to the final algorithm:

• E1 (grayscale + Otsu) found most egg pixels but also bright backgrounds and fingers: recall 0.87, precision 0.38. This showed brightness alone is not enough.

• E2 (HSV colour rule) used saturation to drop coloured things; precision rose to 0.55 but dim eggs were lost.

• E3 added opening (kernel 9). Closing reduced the score in every setting (it merged eggs with bright background) and was dropped.

• E4 added a blob filter. The border-contact filter hurt in every setting because close-up eggs touch the border, so it was switched off. E4 also removed some real eggs: touching eggs form long blobs.

• E5 (watershed) lowered the score and was discarded; the best setting was the one that splits least, so over-splitting is the likely reason (not verified).

• E6 raised the maximum blob aspect ratio from 2.5 to 3.0 after diagnosing two images whose egg blobs had aspect 2.7 to 3.6. Tuning stopped here to avoid overfitting the development set.

4. Task 3: evaluation against the pseudo-ground truth

The SAM masks are treated as reference and the classical masks as predictions. Per image I compute IoU (overlap divided by union), Dice, precision and recall on binary masks. Empty-mask rules are fixed: if both masks are empty the score is 1; precision of an empty prediction is 1 only when nothing was expected; recall is 1 when nothing was expected. **Macro** averages are the mean of per-image scores (every image counts equally); **micro** averages pool the pixels of all images (large eggs and large false regions count more). The evaluation set was used once, after the algorithm was frozen.

Metric	Development (88)	Evaluation (22)
Macro IoU	0.496	0.409
Macro Dice	0.618	0.525
Macro precision	0.684	0.577
Macro recall	0.677	0.647
Median IoU	0.537	0.475

Metric	Development (88)	Evaluation (22)
Micro IoU	0.400	0.280
Micro Dice	0.571	0.437
Micro precision	0.496	0.337
Micro recall	0.674	0.622

Egg counting on the evaluation set: the number of predicted blobs equals the reference count in 12 of 22 images (mean absolute error 0.64 eggs). The evaluation score is lower than the development score; with only 22 images and a large spread (IoU standard deviation 0.27) this gap is consistent with mild overfitting to the development set and with sampling noise, and should not be over-interpreted.

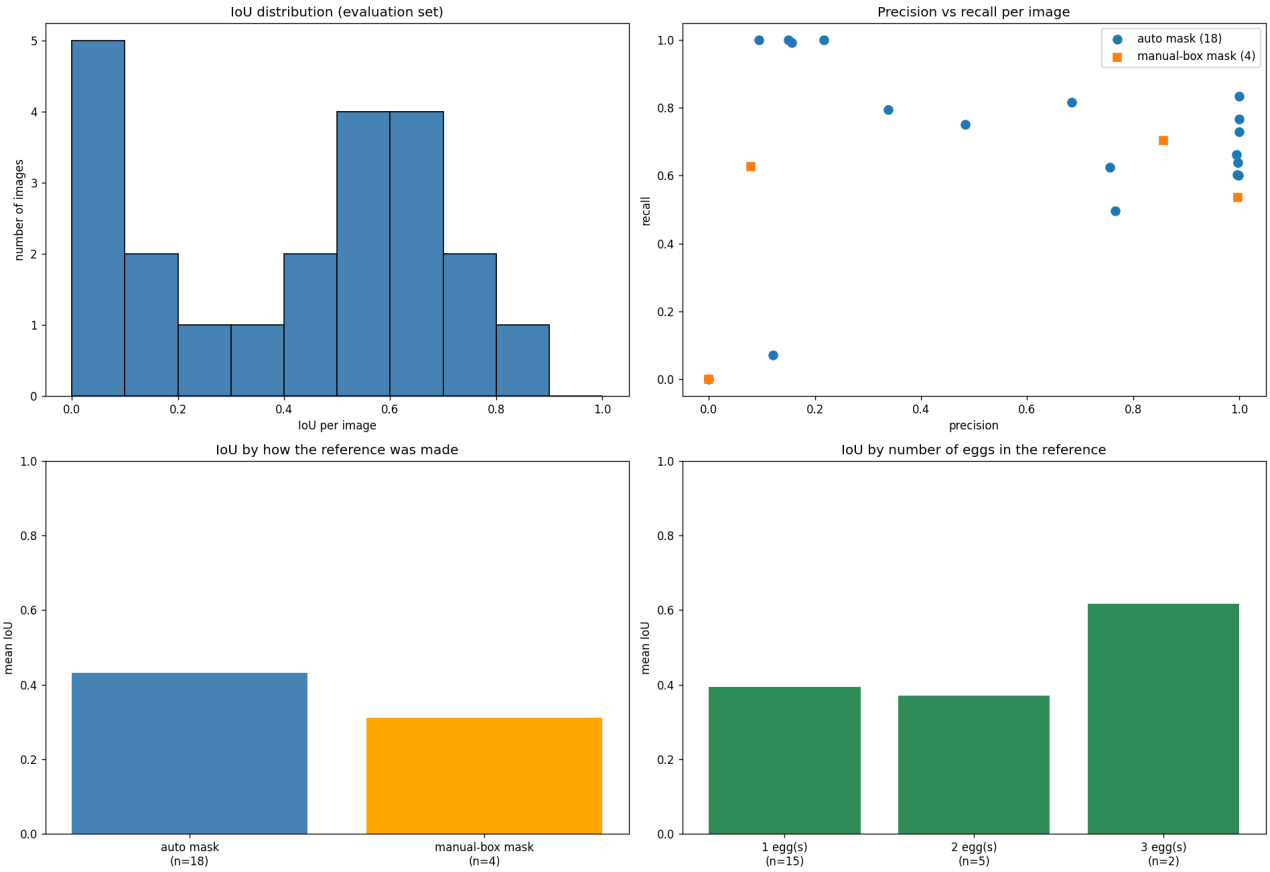


Figure 1: IoU distribution, precision against recall, IoU by reference type (automatic or box), and IoU by number of eggs (evaluation set).

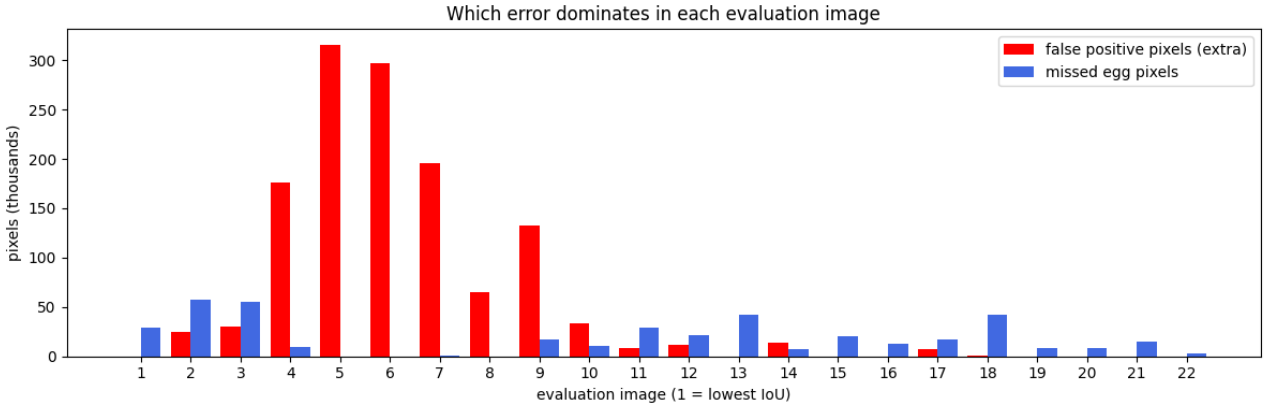


Figure 2: false-positive pixels (red) and missed egg pixels (blue) for each evaluation image, sorted from the lowest IoU. It shows which error type dominates in each image.

Worst six evaluation images:

Image	Reference type	Ref eggs	Pred eggs	IoU	Precision	Recall
not_damaged_117_jpg.rf.d	auto_strict_plus_box	2	0	0.00	0.00	0.00
not_damaged_78_jpg.rf.d3	auto_relaxed	1	2	0.00	0.00	0.00
not_damaged_27_jpg.rf.9e	auto_relaxed	1	3	0.05	0.12	0.07
not_damaged_42_jpg.rf.66	manual_box	1	2	0.08	0.08	0.63
not_damaged_104_jpg.rf.3	auto_strict	1	1	0.09	0.09	1.00
not_damaged_111_jpg.rf.d	auto_strict	1	1	0.15	0.15	1.00

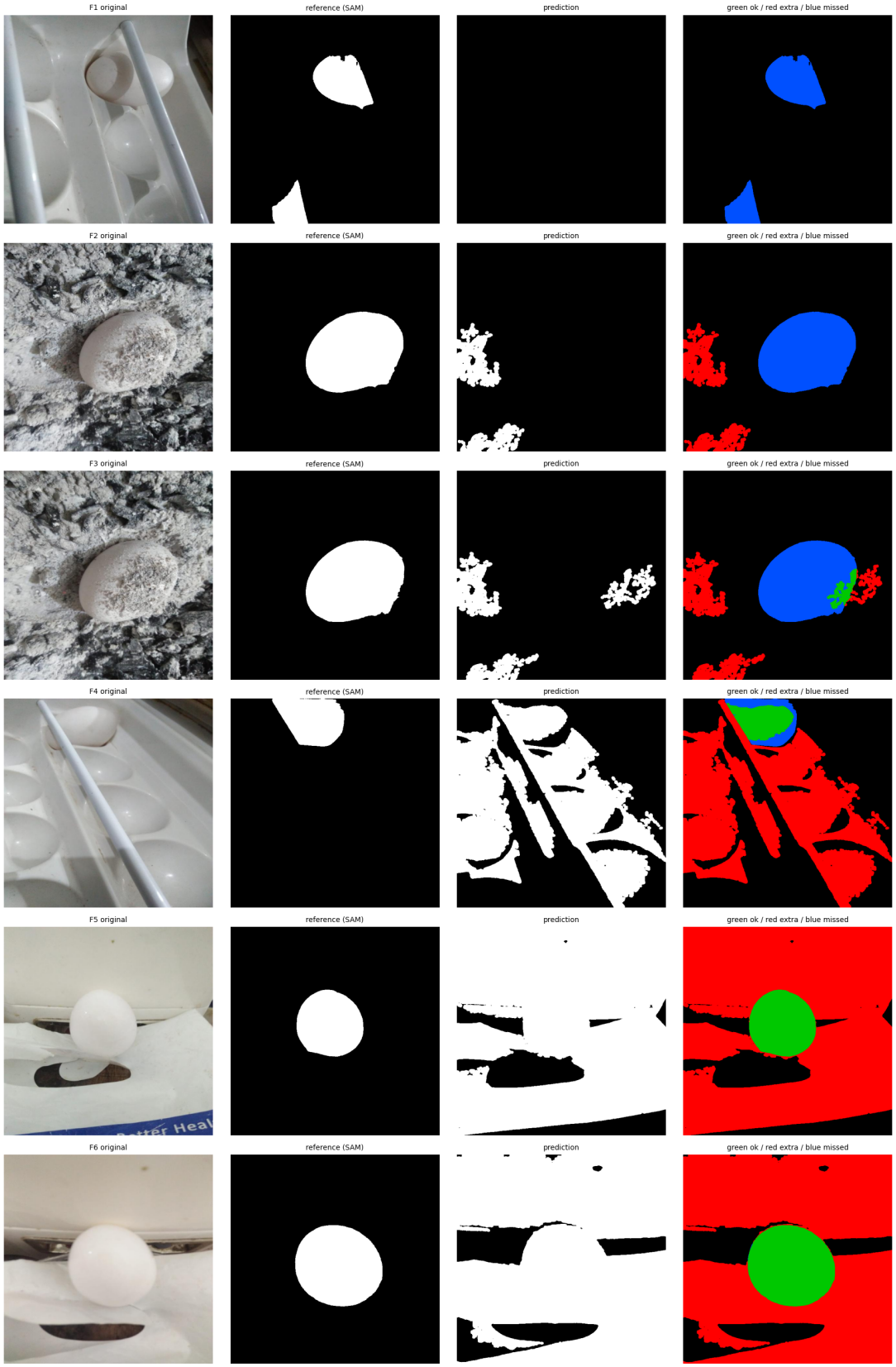


Figure 3: the six worst evaluation images (original, reference, prediction, error map: green correct, red false positive, blue missed).

Qualitative analysis

• Missed or partly missed eggs (blue): the brightness rule drops the shadowed side of an egg and dim or translucent eggs, which lowers recall to about 0.65. One evaluation image has IoU 0 with no egg predicted

at all.

• Large false positives (red): images with recall near 1 but precision around 0.1 mean the prediction covers the egg plus a much larger region. Colour alone cannot separate an egg from a white tray, panel or wall of the same colour; this appeared in development as well (a white panel behind an egg).

• Count errors: touching eggs can merge into one blob (under-segmentation) while textured backgrounds or tray pockets can add extra blobs (over-segmentation); 10 of 22 evaluation images have the wrong blob count.

• Shadows and reflections: closing and hole filling help only for small gaps; the dim side of an egg is the dominant shadow effect.

• Reference errors: in several development error maps the 'missed' (blue) pixels were fingers or tile patterns that SAM had marked as egg. In those cases the classical method is closer to the truth than the score suggests, so the true accuracy is not measured by these numbers.

• Potential improvements: brightness relative to the local background or a Lab colour space for shadowed eggs; filling each egg blob with its convex hull; a gentler watershed; verifying a subset of the reference masks by hand.

5. Task 4: video segmentation and tracking

Video: 854x480, 24 FPS, 60.1 s, 1443 frames. Eggs are visible from about 10 s to 41 s (the rest is text slides), a hand moves and lifts the eggs, and the video contains 2 eggs in total. Only 10 s to 41 s (744 frames) was processed; every frame was used.

Segmentation. The dataset settings found only part of the eggs in the video (video eggs are darker). I tried three video-only versions, evaluated against ovals drawn inside boxes I marked on 4 frames (14, 20, 32 and 34 s) and checked visually on 16 preview frames (cards below):

Trial V1	**Video only: retune HSV saturation/brightness limits on 4 frames (oval proxy from manual boxes); opening and blob filter unchanged**
Why tried	Video eggs are darker/warmer than the dataset eggs, so the dataset thresholds miss them. Preview showed missed eggs in about half of the sampled frames (18, 20, 32, 34-40 s).
Parameters	{'sat_max': 40, 'val_min': 120}
Result	proxy IoU 0.3037 on 4 hand-boxed video frames (not comparable with E1 to E6)
Decision	discard
Observations / next	Looser colour limits recovered some eggs but also marked shirts and pants; colour alone cannot separate eggs from pale or grey clothing. Next: V2: restrict to a horizontal tracking zone where eggs were seen and drop oversized blobs

Trial V2	**Video only: V1 colour sweep + tracking zone y=160-440 + max blob size 12% of frame; opening and blob filter as E6**
Why tried	Eggs stay in a horizontal zone and are smaller than clothes; position and size rules remove clothing that colour cannot. V1 marked shirts and pants.
Parameters	{'sat_max': 100, 'val_min': 120} zone=160-440 max_area=0.12
Result	proxy IoU 0.4431 on 4 hand-boxed video frames (not comparable with E1 to E6)
Decision	discard (table still marked)
Observations / next	Brightness limit is the key setting (proxy IoU 0.20 at 170 vs 0.44 at 120). Zone + size limit removed shirts/pants, but table still marked at 22-34 s (blobs with brightness 111-163, solidity 0.55-0.79) and one egg missed at 36 s. Next: V3: bright core (eggs are brightness >200, table is dimmer) then small grow

Trial V3	**Video only: bright, low-saturation core -> opening -> small dilation -> keep only where loose rule (sat<=100, val>=120) agrees; zone y=160-440; blob filter min 1%, aspect 3.0**
Why tried	Eggs have a bright core (brightness >200); table pieces are dimmer, so a core-based rule drops the table. V2 still marked the table (blobs val 111-163, solidity 0.55-0.79) and merged eggs with table.
Parameters	{'core_val': 170, 'grow': 21}
Result	proxy IoU 0.5538 on 4 hand-boxed video frames (not comparable with E1 to E6)
Decision	keep, final video segmentation
Observations / next	Bright core (sat<=60, val>=170) + opening + 21 px grow, limited to loose rule (sat<=100, val>=120), zone y=160-440. Proxy IoU 0.554 on 4 boxed frames (V2 0.443); right egg count in 3 of 4 frames; 16-frame preview judged good by visual check. Next: Build centroid tracker with persistent IDs and render annotated video

The final video version V3 finds a bright low-saturation core (saturation at most 60, brightness at least 170), applies opening, grows it by 21 pixels, and keeps pixels only where a looser colour rule agrees, inside the zone (image rows 160 to 440). These video settings, including the fixed zone, were tuned on 4 frames and are specific to this video.

Tracking. For each blob I compute the centroid, bounding box and outline. Tracks and blobs are matched greedily by distance (closest pairs first, at most 100 px, plus 5 px per frame a track was unseen). A new track receives an ID after 3 frames (eggs visible in the first frame get one at once), a lost track is kept for 24 frames (1 s) so a returning egg keeps its ID, and a new blob within 40 px of an already matched blob is ignored to avoid duplicate IDs. The output video shows outlines, boxes, centroid trails and 'egg N' labels.

Results on the full run: tracked eggs per frame - 2 eggs in 506 frames, 1 egg in 231 frames, 0 in 7 frames. Eight IDs were created for the 2 real eggs:

ID	Frames seen	First (s)	Last (s)	Comment (my reading of the video)
1	225	10.0	19.3	first egg until it was lifted
2	494	10.0	31.0	long track; at the start the two close eggs formed one blob
3	3	21.0	21.1	egg lifted by the hand, then hidden
4	96	23.2	27.2	egg put back after being hidden for more than 1 s: new ID
5	1	28.3	28.3	one-frame noise
6	7	28.6	28.8	short noise
7	244	30.8	41.0	egg put back again: new ID
8	173	33.8	41.0	egg re-detected after the hand moved the eggs: new ID

Failure modes: an egg hidden or lifted for more than 1 second gets a new ID on return (identity is not preserved across hand occlusions); two close eggs can merge into a single blob and share one ID; tracks of one to seven frames are noise. A possible improvement is to keep lost tracks longer and ignore very short tracks; I did not have time to test it.

Annotated video (10 s to 41 s): https://drive.google.com/file/d/1nzwb-UZDESOID0dqhEN9Ej4_-i6Tr7pG/view?usp=sharing. I verified that the link plays in a private browser window without signing in.

6. Limitations

• The reference is SAM output, partly steered by my manual boxes (17 of 110 images). It is not human ground truth, and SAM's own errors count against the classical method. The scores measure agreement, not accuracy.

• The SAM selection rule was tuned by inspecting images from the whole dataset, including images that later fell into the evaluation set. The classical algorithm was tuned on the development set only.

• The evaluation set is small (22 images), so scores carry large uncertainty. The perceptual-hash grouping found no near-duplicates, so the split is effectively a seeded random split.

• A threshold-based method cannot handle dim, shadowed or translucent eggs, or backgrounds with the same colour as the eggs. Watershed did not help (over-splitting is a likely but unverified cause).

• Video settings were tuned on 4 frames with rough oval proxies and use a fixed zone; ID continuity breaks when a hand lifts or hides an egg.

• The src/ code in the repository is a cleaned copy of the notebook logic that was smoke-tested on synthetic data only; the notebook produced all reported numbers.

7. Setup and reproducibility

• Repository: https://github.com/sarangp17/egg-segmentation-tracking (README has setup commands, dataset download, notebook cell order, parameters in configs/final_params.json).

• Environment: Google Colab; SAM needs a GPU (T4 used), everything after the reference masks runs on CPU. Packages are listed in requirements.txt. SAM ViT-B checkpoint is downloaded separately (not in the repository).

• Dataset: download version 1 in 'folder' format with a (personal) Roboflow API key; images are not redistributed. The source video is downloaded separately.

• Seeds and parameters: split seed 42; every frozen parameter is in configs/final_params.json; every trial is in experiments/experiment_log.csv; the split is in experiments/split_manifest.csv.

8. AI-tool disclosure

I used Claude (Anthropic) throughout this project to plan the work, propose methods and parameter sweeps, write and debug code, and draft this report and the README. Claude wrote most of the code. My own work: I set up and ran every notebook in Google Colab; checked each stage visually (SAM overlays, error maps, video previews); supplied the manual box prompts for 17 of the 110 reference masks and the boxes used to tune the video settings; reported the failures I saw (for example clothes and the table marked as eggs, missed eggs), which led to changes in the method; and reviewed and approved the keep/discard decision for each experiment. Every number in this report comes from runs I executed. SAM is a pretrained model by Meta AI (Apache-2.0).

Credits: dataset 'egg' by a Roboflow Universe user (fire-bf7qe), CC BY 4.0; SAM by Meta AI (Apache-2.0); source video on YouTube (not redistributed).